

DETAILED Research Plan

GOAL: littoral reconnaissance, collision avoidance for autonomy, small-craft detection under clutter

Phase 1: Foundation & CNN Baseline (Oct 2025 – Feb 2026)

Train completed CNN pipeline and begin training initial self-supervision models.

Tasks & Goals:

- **Data collection:**
 - Nathan and I lead collection of littoral radar data across various sites
 - Sam builds tracking baselines, not detection/classification which is what the CNN will do.
 - Target 2,000 labeled frames by Feb 2026 for supervised training, currently have 200,000 unlabeled frames
 - Reduce human labelling time via better pre-labelling; proposed drop from 60s to 20s per frame by reducing missed objects.
- **Model development:**
 - Train CNN object detector (YoloV8/Radar-variants) with (assisted-labelling pipeline)**need better name here for the idea that we can use GPS and heading info and traditional algorithms to pre-label all images with bounding boxes – like 80% accuracy**;
 - Train self-supervised model targeting within 15% of supervised baseline.
- **Paper writing:**
 - IEEE paper on CNN baseline performance on Furuno Radars - ME
 - AIAA papers on CFAR parameter tuning - NATHAN
 - AIAA on DB-SCAN and ST-DBSCAN object tracking - SAM
 - AIAA paper(s) on HDR for pulse and gain as well as an entropy piece – ME or UNDERGRADUATES
 - Paper on (assisted-labelling) methodology vs. manual labelling.
- **Team:**
 - Sam builds tracking via ST-DBSCAN and traditional methods, including Kalman.
 - Nathan handles data collection and CFAR parameter tuning
 - Recruit 2 undergraduates to begin training them for hardware/software project phases
 - Radar-assisted camera models – software
 - Camera-assisted radar models – software
 - Multi-spectrum camera setup and models – hardware and software
 - Slew-to-cue with zoom – hardware and software
 - Autonomous boat system integration – hardware only rn

Phase 2: Multi-Modal Scaling (Feb 2026 – Aug 2026)

Transition to hybrid radar-camera models and expand dataset dramatically.

Concurrent tasks:

- **Leadership:**
 - Transition Sam to lead CNN + self-supervised radar models as well as continuing classic tracking
 - Nathan focuses on radar-camera calibration and documentation
 - Nathan begins training undergraduate for data collection (Nathan is graduating!!)
 - Take over data capture from other teams on project to streamline research efforts – i.e. multi-spectrum camera, radar, stereo camera, lidar all get handled during single trip instead of each team doing separate trips, reduces cost.
 - Begin submitting funding proposals for all undergraduates that are not doing internships

- **Data:**
 - Increase labeled frames to 5,000 by May – only 50 man-hours needed with 60s per frame
 - Increase labelled frames to 10,000 by (Aug) – only 85 man-hours
 - begin undergraduate recruitment from engineering 101 classes – freshmen make the best researchers because they are interested in doing something cool, and I've got the resources
 - build camera-specific models (RGB-only tracking) for auto-labelling comparison
- **Model development:**
 - Build Radar-Assisted vision models (i.e. using radar to cue camera models)
 - Build Camera-Assisted radar models (i.e. using camera to correct radar classification)
 - Build combined multi-modal hybrid models using radar and camera together
 - Continue to improve self-supervised methods to close gap to baseline.
- **Papers & proposals:**
 - Write paper comparing auto-labelling methods and my assisted-labelling pipeline combined with Sam's work on labelling
 - Submit McNair Junior Fellowships (or alternative) for each undergraduate seeking to do work
 - Create research posters for ALL undergrads that completed work over summer for Summer Research Symposium
- **Infrastructure:**
 - Completely build out radar + camera system on pontoon boat system for prototyping and testing – also needs undergrads for this
 - Complete data collection pipeline on littoral autonomous boat
 - Validate real-time end-to-end tracking latency – may require rewrite/refactor of entire portions of code base

Phase 3: Slew-to-Cue & Model Hardening (Aug 2026 - May 2027)

Deploy slew-to-cue capability and release Dataset v0.

Concurrent tasks:

- **Feature development:**
 - Move slew-to-cue setup to autonomous boat
 - Should be able to start ramping up data collection due to 'autonomous' nature.
 - Also, should allow for beginning RL work on autonomous marine vehicles.
 - Validate closed-loop detection-to-track latency < 500ms.
- **Dataset:**
 - Target 20,000 human-labelled frames by May '27
 - Initial release of Dataset v0 (radar data labelled)
 - Labelled camera data is released by separate team – likely already at v1 or v2 at this point.
 - Categorize models into non-moving/littoral vs. slow-moving vs. fast-moving classes
 - Fast moving means the radar is moving at more than 10 mph – more reliance on AIS/GPS
 - Slow-moving under 10 mph – much easier to label and infer object type from
- **Team expansion:**
 - Recruit more undergrads for additional phases of project
 - 1-2 undergrads: Need to start doing multiple data captures simultaneously – i.e. one in Charleston and one in lake (lake = early/prototype models and testing; coast = almost-ready for production; open harbor = fully ready for deployment)
 - 1-2 undergrads: multi-spectrum system design, gimbals for it.
 - 2-3 undergrads: model development & multi-modal auto-labelling system dev
- **Model refinement:**
 - Complete training of slow-moving models
 - Re-trained littoral model with motion augmentation
 - Implement moving models on small perception boats for dockside testing

- Use this for additional data collection as stated prior
- **Papers:**
 - Write paper on slew-to-cue integration
 - write paper on radar-assisted + camera-assisted fusion method
 - Submit proposals for government collaboration: NIWC, NOAA

Phase 4: Multi-Spectral Integration (May 2027 – December 2027)

Deploy thermal (FLIR) and short-wave IR cameras; expand fleet

Concurrent tasks:

- **Engineering:**
 - Mechanical for multi-spectral gimbals
- **ML development:**
 - Build multi-spectral models; takes advantage of multiple input data sources to create best 'world model' for obstacle avoidance and crash reductions.
 - Fine-tune hybrid model on synthetic multi-spectral data – relies on RL
 - Distill 100B-param model to real-time deployable version
 - Use labelled data from 100B model to train lightweight supervised models
 - Publish paper comparing littoral vs moving models in a moving environment
 - Document various labelling pipelines across team,
 - Publish flagship papers – May '28
- **Team scaling:**
 - Push for team leads to acquire further undergraduate help
- **Fleet ops:**
 - Deploy 1-3 autonomous boats in lakes and coastal waters
 - Reinforcement learning models to flag uncertain frames for human review
- **ML improvement:**
 - Begin long-term temporal tracking on slow-moving auto-boats (hours).
- **Large vessel track:**
 - Install sensor suite on 2 NOAA ships;
 - integrate with NOAA biologists' annotations;
 - validate on targets beyond littoral clutter.

Phase 7: Transition to Production (May 2028)????

- *Commercial:* Engage shipping/fisheries partners
- *Defense:* Scope naval use cases (UAS swarms, periscope detection); pursue CRADA? What is dis?
- *Science:* Package model-as-a-service for marine biologists; integrate with NOAA data portals